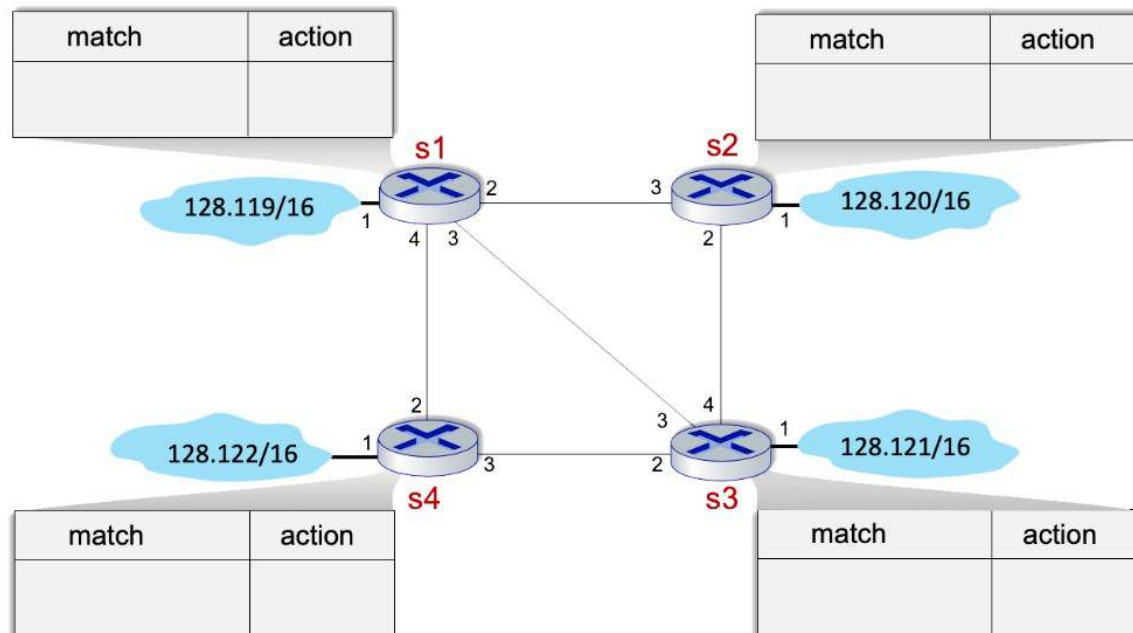


Consider the 4-router network shown below, where packet forwarding is controlled by flow tables (e.g., configured via OpenFlow in an SDN controller), rather than by a forwarding table computed by a routing algorithm. The addresses of networks attached to each of the router is also shown. The interfaces at each of the routers are also as indicated.



Suppose we want the following forwarding behavior of packets to be implemented:

- Packets coming from the source network attached to s3 and destined to the network attached to s2 should be forwarded along the path: s3 -> s4 -> s1 -> s2. Assume that at s3 that only HTTP traffic is allowed to exit the network.

Complete the match-plus-action tables in each of the routers, s1, s2, s3, and s4, that implement these forwarding behaviors. Your rules should be as strict as possible (should only allow these behaviors, and no other forwarding behaviors). You can assume that any packet arriving at a router that does not match a rule in that table will be dropped.

Answers

(Q1-Q7) Rule 1. | IP_Src=128.121/16, IP_Dst=128.120/16, Src_Port=Any, Dst_Port=Any, IP_Protocol=Any, Action=Forward(2) |

(Q8-Q14) Rule 2. | IP_Src=128.121/16, IP_Dst=128.120/16, Src_Port=Any, Dst_Port=Any, IP_Protocol=Any, Action=Forward(1) |

(Q15-Q21) Rule 3. | IP_Src=128.121/16, IP_Dst=128.120/16, Src_Port=Any, Dst_Port=80, IP_Protocol=TCP, Action=Forward(2) |

(Q22-Q28) Rule 4. | IP_Src=128.121/16, IP_Dst=128.120/16, Src_Port=Any, Dst_Port=Any, IP_Protocol=Any, Action=Forward(2) |

CRC

ERROR DETECTION AND CORRECTION: CYCLIC REDUNDANCY CHECK

Consider the Cyclic Redundancy Check (CRC) algorithm discussed in Section 6.2.3 of the text. Suppose that the 4-bit generator (G) is 1001, that the data payload (D) is 10011011 and that $r = 3$.

Solution

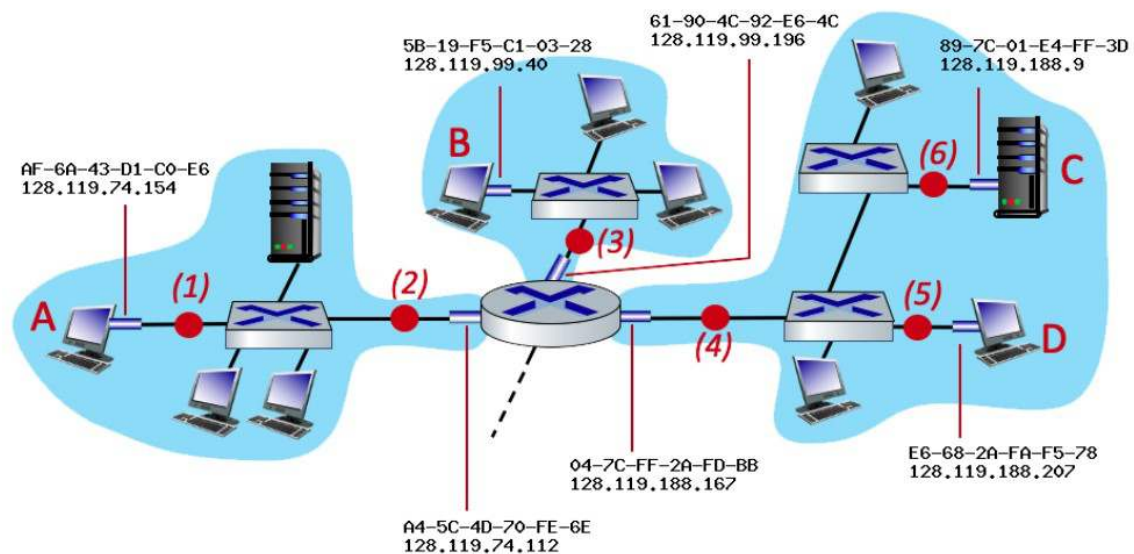
To compute the CRC, we begin by taking the value of D, 10011011, and multiplying it by 2^3 , giving 10011011000. We then divide this number by the generator bits [G] = 1001, using modulo-2 arithmetic. The final remainder, R, after this division are then the CRC bits. Here is that calculation:

$$\begin{array}{r} 10001010 \\ \hline 1001 \) \ 10011011000 \\ \underline{1001} \\ 1011 \\ \underline{1001} \\ 1000 \\ \underline{1001} \\ 010 \rightarrow R \end{array}$$

We've computed the remainder as $R = 010$ and the quotient $n = 10001010$. You should verify that $n * G \text{ XOR } R$ is indeed equal to 10011011000. You can use [this calculator](#) to do the modulo-2 arithmetic if you don't want to do it by hand.

LINK LAYER (AND NETWORK LAYER) ADDRESSING AND FORWARDING

Consider the figure below. The IP and MAC addresses are shown for nodes A, B, C and D, as well as for the router's interfaces.



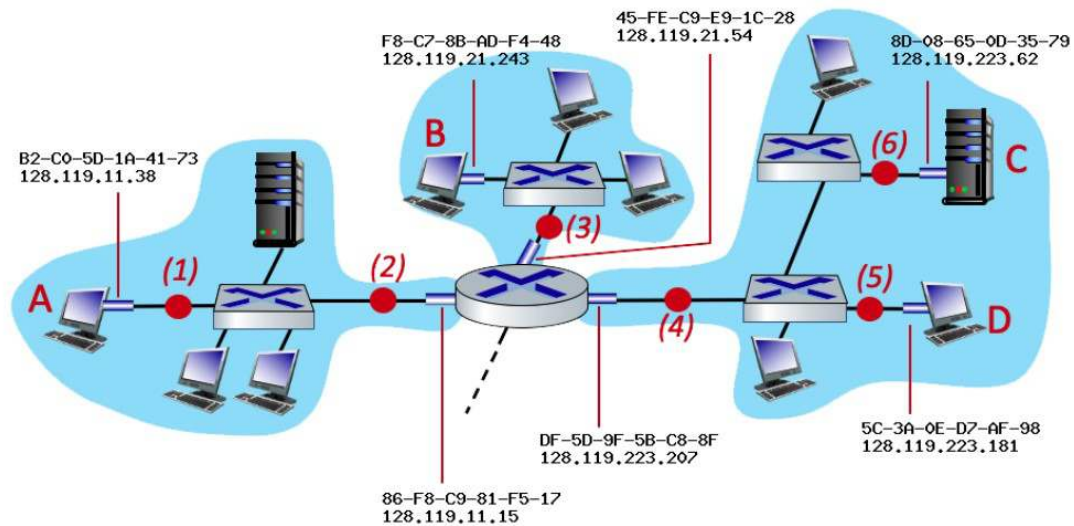
Consider an IP datagram being sent from node C to node D.

SOLUTION

1. The source mac address at point 6 is 89-7C-01-E4-FF-3D
2. The destination mac address at point 6 is E6-68-2A-FA-F5-78
3. The source IP address at point 6 is 128.119.188.9
4. The destination IP address at point 6 is 128.119.188.207
5. No, datagrams can be sent across the subnet via the link layer in one go.

Q2.

Consider the figure below. The IP and MAC addresses are shown for nodes A, B, C and D, as well as for the router's interfaces.



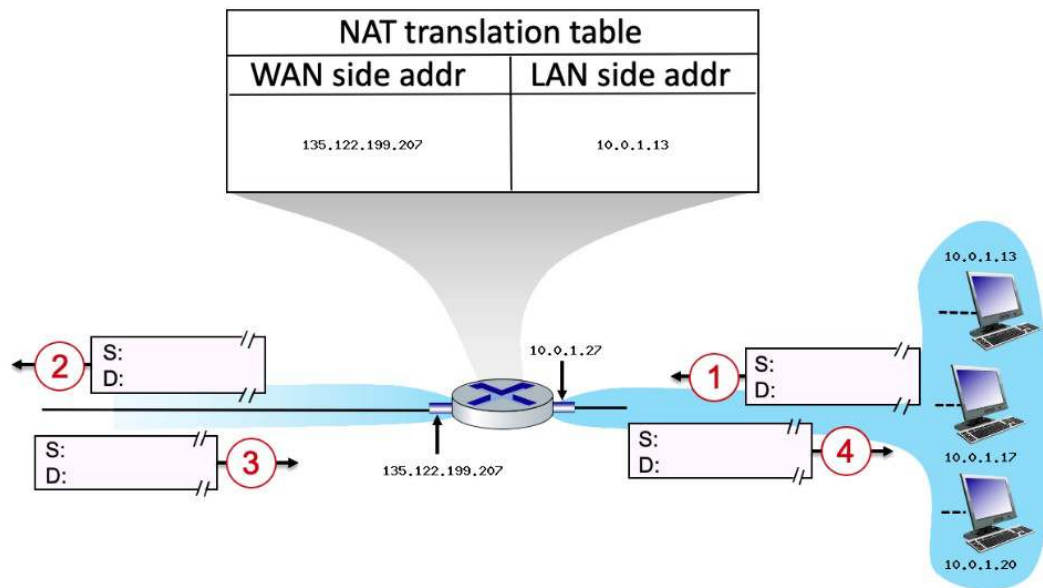
Consider an IP datagram being sent from node **A** to node **B**.

SOLUTION

1. The source mac address at point **1** is B2-C0-5D-1A-41-73
2. The destination mac address at point **1** is 86-F8-C9-81-F5-17
3. The source IP address at point **1** is 128.119.11.38
4. The destination IP address at point **1** is 128.119.21.243
5. No, datagrams can be sent across the subnet via the link layer in one go.
6. Yes, the datagram is being passed through a router, which causes a subnet change.
7. The source mac address at point **3** is 45-FE-C9-E9-1C-28
8. The destination mac address at point **3** is F8-C7-8B-AD-F4-48

NAT

Consider the scenario below in which three hosts, with private IP addresses 10.0.1.13, 10.0.1.17, 10.0.1.20 are in a local network behind a NAT'd router that sits between these three hosts and the larger Internet. IP datagrams being sent from, or destined to, these three hosts must pass through this NAT router. The router's interface on the LAN side has IP address 10.0.1.27, while the router's address on the Internet side has IP address 135.122.199.207



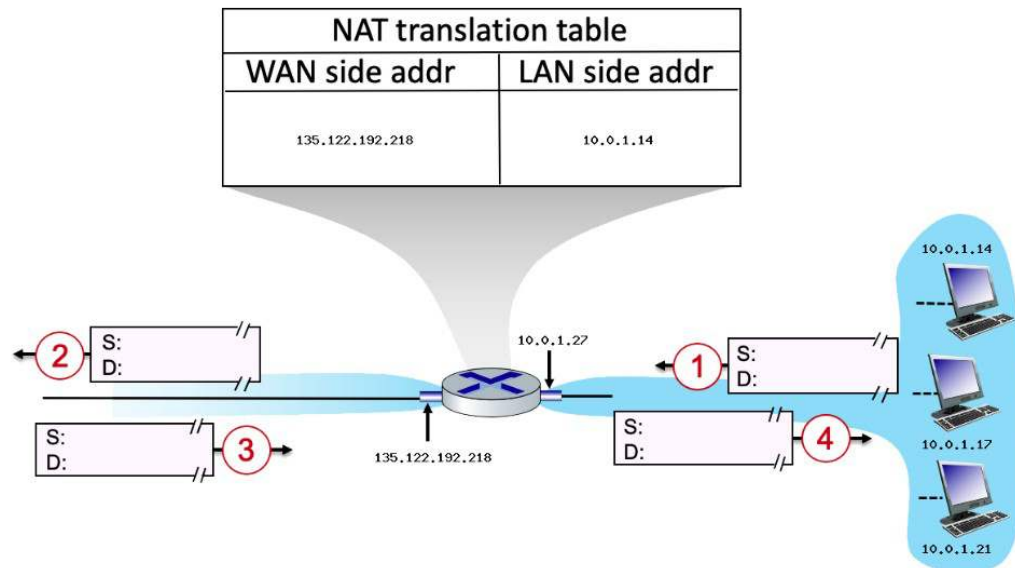
Suppose that the host with IP address 10.0.1.13 sends an IP datagram destined to host 128.119.169.189. The source port is 3390, and the destination port is 80.

SOLUTION

1. The source address will be the local host's IP, which is 10.0.1.13
2. The destination address will be the remote machine's IP, which is 128.119.169.189
3. The source address will be the router's public IP, which is 135.122.199.207
4. The destination address will be the remote machine's IP, which is 128.119.169.189
5. Yes, the NAT will change the source port.
6. The source address will be the remote machine's IP, which is 128.119.169.189
7. The destination address will be the router's public IP, which is 135.122.199.207
8. The source address will be the remote machine's IP, which is 128.119.169.189
9. The destination address will be the local host's IP, which is 10.0.1.13
10. No, an entry is made when there's an outbound request, which only happens between step 1 and step 2.

Another Practice Problem

Consider the scenario below in which three hosts, with private IP addresses 10.0.1.14, 10.0.1.17, 10.0.1.21 are in a local network behind a NAT'd router that sits between these three hosts and the larger Internet. IP datagrams being sent from, or destined to, these three hosts must pass through this NAT router. The router's interface on the LAN side has IP address 10.0.1.27, while the router's address on the Internet side has IP address 135.122.192.218.



Suppose that the host with IP address 10.0.1.14 sends an IP datagram destined to host 128.119.162.185. The source port is 3400, and the destination port is 80.

SOLUTION

1. The source address will be the local host's IP, which is 10.0.1.14
2. The destination address will be the remote machine's IP, which is 128.119.162.185
3. The source address will be the router's public IP, which is 135.122.192.218
4. The destination address will be the remote machine's IP, which is 128.119.162.185
5. Yes, the NAT will change the source port.
6. The source address will be the remote machine's IP, which is 128.119.162.185
7. The destination address will be the router's public IP, which is 135.122.192.218
8. The source address will be the remote machine's IP, which is 128.119.162.185
9. The destination address will be the local host's IP, which is 10.0.1.14

10. No, an entry is made when there's an outbound request, which only happens between step 1 and step 2.